

1. A scalar field ϕ and a pseudoscalar field π interact with electrons and positrons through the following interaction: $L = g\bar{\psi}\psi\phi + ig'\bar{\psi}\gamma_5\psi\pi$. Calculate the differential cross section $\frac{d\sigma}{d\Omega}$ averaged over initial spins and summed over final spins for the process $e^- + \phi \rightarrow e^- + \pi$ to lowest nontrivial order in the lab frame. You may take the ϕ and the π to be massless.
2. Compute the differential cross section $\frac{d\sigma}{d\Omega}$ averaged over initial spins and summed over final spins for the process $e^+ + e^- \rightarrow \mu^+ + \mu^-$ in to lowest nontrivial order in Quantum Electrodynamics in the center of mass frame. Note that the muons couple to the photons in the same way as the electrons (why?).